Project Report: Heart Disease Analysis using Tableau

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1. PROJECT TITLE

Heart Disease Analysis using Tableau

2. PROJECT DESCRIPTION

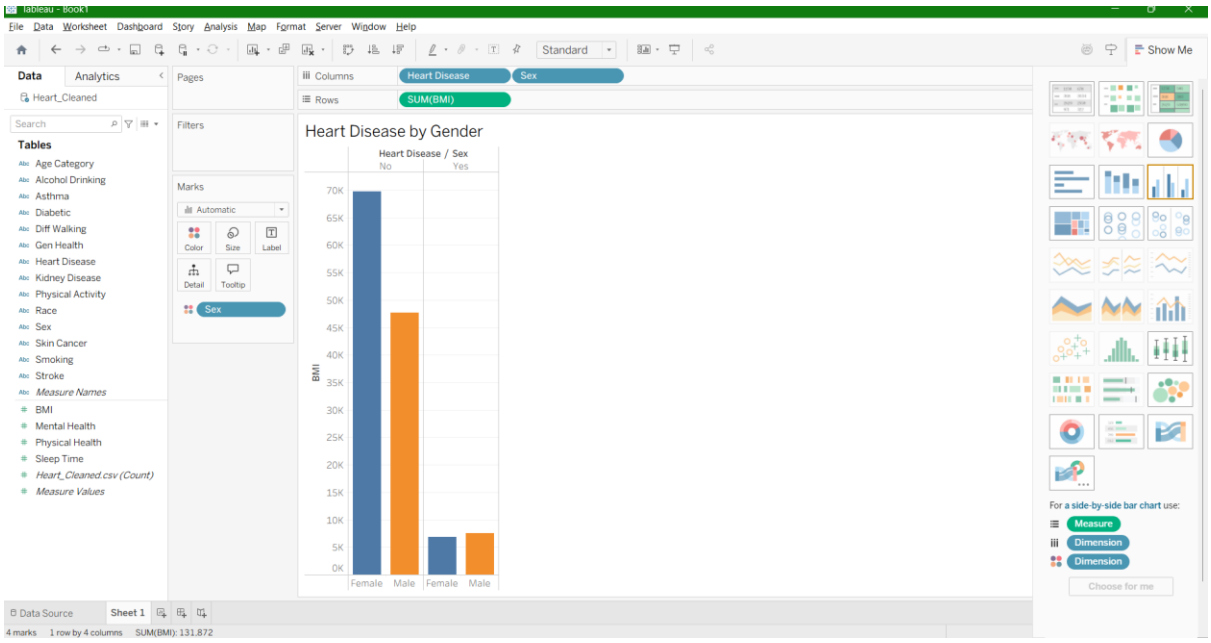
- **System Overview:** A Tableau-based analytical system for visualizing heart disease risk factors.
- **Problem Domain:** Cardiovascular health analysis and preventive healthcare.
- **Core Technology Used:** Tableau, CSV datasets, data cleaning tools.
- **Key Capabilities:** Interactive dashboards, comparative analysis, storytelling with data.
- **Target Users:** Healthcare professionals, researchers, students.
- **Real-world Relevance:** Helps identify lifestyle and demographic factors contributing to heart disease.

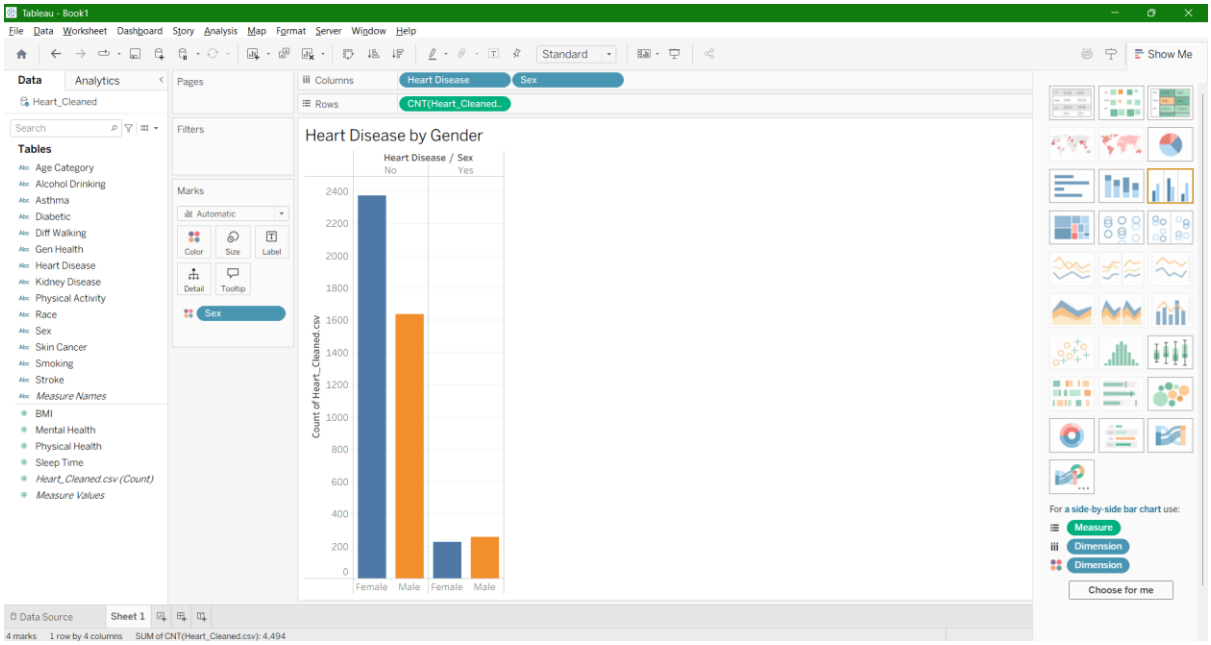
3. APPLICATION SCENARIOS / USE CASES

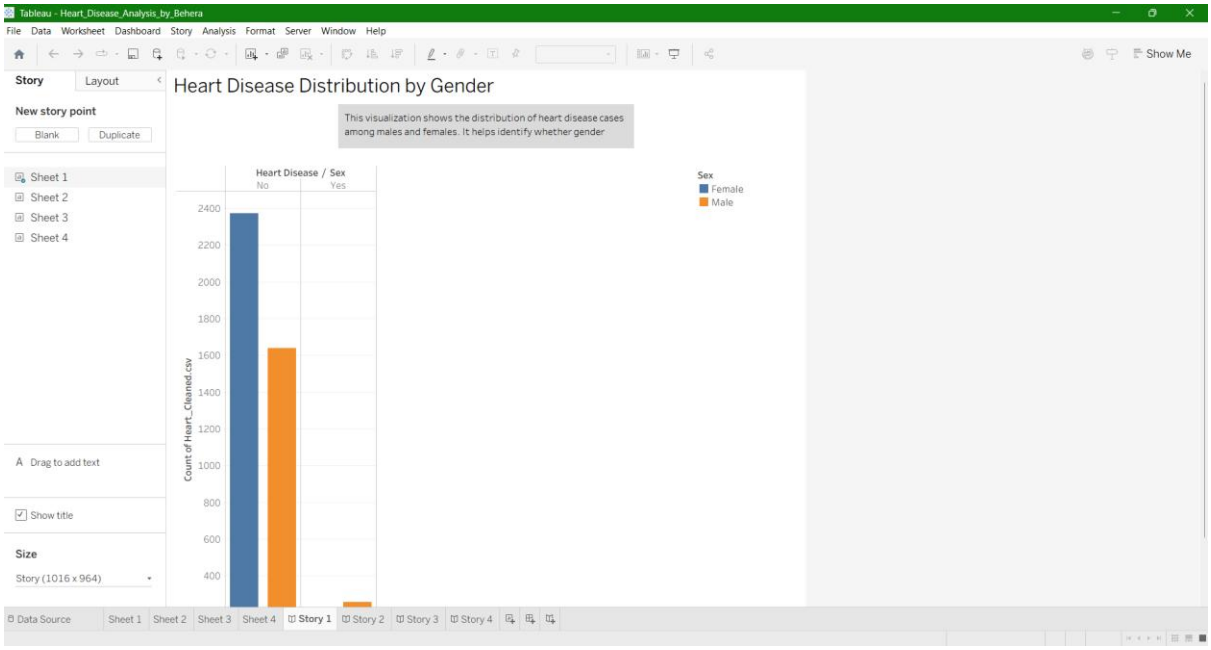
- **Healthcare Institutions:** Use dashboards to monitor patient risk factors.
- **Educational Usage:** Teaching students about data visualization and health analytics.
- **Research:** Supporting studies on lifestyle impacts on cardiovascular health.
- **Public Awareness:** Visual stories for campaigns on preventive healthcare.

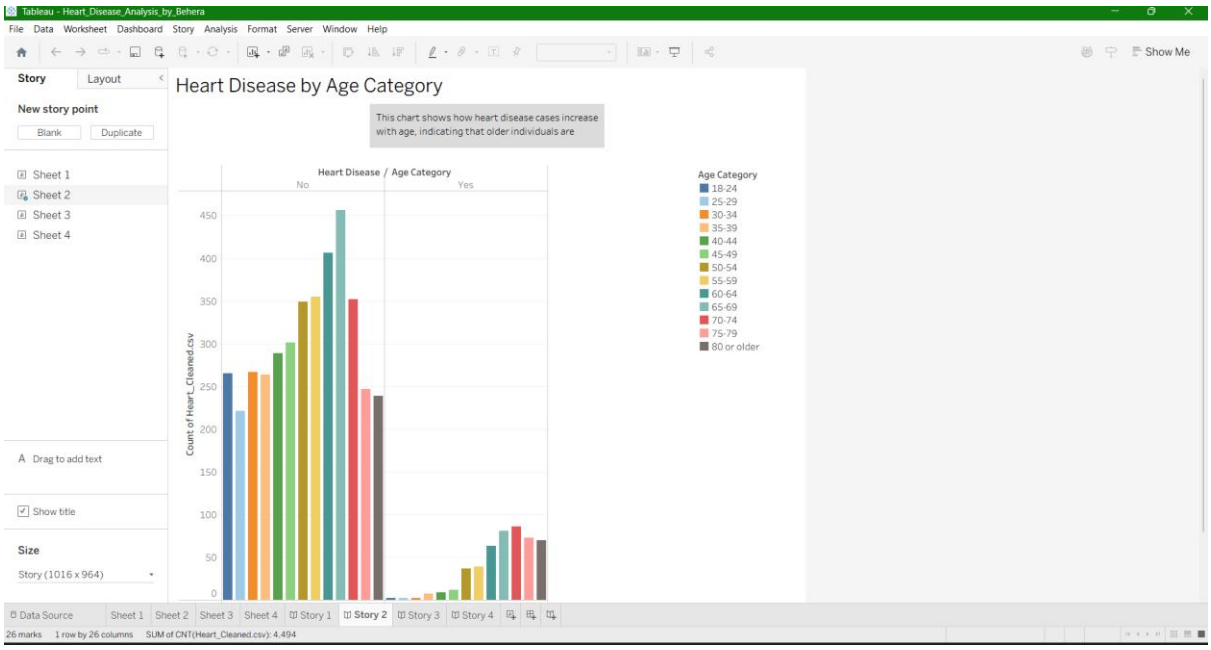
4. TECHNICAL ARCHITECTURE OVERVIEW

- **Frontend Layer:** Tableau dashboards and storyboards.

- **Backend Layer:** CSV dataset preprocessing.
- **Database Layer:** Heart_Cleaned.csv file.
- **External APIs/Services:** None.
- **Deployment Environment:** Tableau Desktop / Tableau Public.
- 

- 

- 

- 

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5. PREREQUISITES

5.1 Software Requirements

- **Tableau Desktop / Tableau Public**
- **Python (optional for preprocessing)**
- **GitHub for version control**

5.2 Libraries / Frameworks / Dependencies

- Pandas (for preprocessing)
- Tableau built-in visualization tools

5.3 Hardware Requirements

- Standard PC/Laptop with 8GB RAM, 2GHz processor

6. PRIOR KNOWLEDGE REQUIRED

- Basics of Tableau
- Data cleaning and preprocessing
- CSV handling
- Visualization principles

7. PROJECT OBJECTIVES

- **Technical Objectives:** Build interactive dashboards.
- **Performance Objectives:** Ensure smooth visualization with large datasets.
- **Deployment Objectives:** Publish dashboards on Tableau Public.
- **Learning Outcomes:** Gain expertise in Tableau storytelling and health data analysis.

8. SYSTEM WORKFLOW

1. User loads dataset into Tableau.
2. Input handling: Cleaned CSV imported.
3. Processing logic: Apply filters, aggregations.
4. Core execution: Generate dashboards.
5. Output generation: Charts, graphs, storyboards.
6. Response delivery: Export reports and share dashboards.

9. MILESTONE 1: REQUIREMENT ANALYSIS & SYSTEM DESIGN

- **Problem Definition:** Identify risk factors of heart disease.
- **Functional Requirements:** Import dataset, visualize variables, create dashboards.
- **Non-Functional Requirements:** Scalability, usability, clarity.
- **System Design Decisions:** Tableau chosen for visualization.
- **Technology Stack:** Tableau + CSV dataset.

10. MILESTONE 2: ENVIRONMENT SETUP & INITIAL CONFIGURATION

- **Development Environment Setup:** Tableau Desktop.
- **Dependency Installation:** Pandas (optional).
- **Project Structure Creation:** GitHub repo with dataset + Tableau files.
- **Configuration Setup:** Folder structure for dataset, dashboards, reports

11. MILESTONE 3: CORE SYSTEM DEVELOPMENT

- **Feature 1:** Age & Gender Analysis (bar charts).
- **Feature 2:** Lifestyle factors (smoking, BMI, physical activity).
- **Feature 3:** Story scenes summarizing findings

12. MILESTONE 4: INTEGRATION & OPTIMIZATION

- Integrated multiple dashboards into a single story.
- Optimized filters for faster response.
- Added validation for dataset consistency.

13. MILESTONE 5: TESTING & VALIDATION

- **Test Cases:** Verified dataset import, dashboard rendering.
- **Unit Testing:** CSV preprocessing.
- **Integration Testing:** Tableau story scenes.
- **User Acceptance Testing:** Team reviewed dashboards.
- **Performance Metrics:** Smooth rendering with ~10k records.

14. DEPLOYMENT

- **Deployment Architecture:** Tableau Public.
- **Hosting Platform:** Tableau Public cloud.
- **Deployment Steps:** Upload dashboards, share link.
- **Production Considerations:** Ensure dataset privacy.

```
project_root/
 │
 ├─── dataset/
 │    └─── Heart_Cleaned.csv
 │
 ├─── tableau_files/
 │    └─── Heart_Disease_Dashboard.twbx
 │
 ├─── reports/
 │    ├─── Behera_Scene_Story-2026.pdf
```

```
│   └── Behera_No_of_Visualizations_or_Graphs-2026.pdf
│
├── documentation/
│   └── Project_Report_DA_Heart_Disease_Analysis.pdf
│
├── images/
│   ├── dashboard_overview.png
│   ├── age_vs_heart_disease.png
│   ├── cholesterol_analysis.png
│   └── chest_pain_analysis.png
│
├── data_processing/
│   └── data_cleaning_notes.txt
│
└── README.md
```

16. RESULTS

- **System Output:** Interactive dashboards.
- **Performance Evaluation:** Efficient visualization.
- **Screenshots:** [Insert Tableau dashboard screenshots].

17. ADVANTAGES & LIMITATIONS

- **Advantages:** Easy visualization, interactive dashboards, real-world relevance.
- **Limitations:** Limited dataset size, no predictive modeling.

18. FUTURE ENHANCEMENTS

- Add predictive ML models.
- Integrate with larger datasets.
- Cloud-based deployment.
- Mobile-friendly dashboards.

19. CONCLUSION

This project successfully demonstrates the use of Tableau for analyzing heart disease risk factors. It highlights the importance of lifestyle choices and provides a foundation for further research and healthcare applications.

20. APPENDIX

- **Source Code:** GitHub repo
- **Dataset Details:** Heart_Cleaned.csv
- **GitHub Repository Link:** https://github.com/SATYAM8126/Heart-Disease-Analysis-Tableau